

# Steven Boada, Ph.D

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| Basic Information       | <div>(615) 200-0119</div> <div>stevenboada@gmail.com</div> <div>github.com/boada</div> <div>linkedin.com/in/theboada</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| Professional Experience | <div><div>Activision Publishing, Inc.</div><div>Boulder, Colorado</div><div>Senior Data Scientist, Gameplay</div><div>January, 2024 – Present</div><div>As part of the game science team, I work to make Call of Duty better through data. Currently, I am:</div><div><ul style="list-style-type: none"><li>Working with senior design leadership to understand how players interact with Call of Duty gameplay systems. This includes maps, modes, movement, weapons, and in-game communication among others.</li><li>Leading initiatives to understand how game design affect topline KPIs, and suggest changes when appropriate.</li><li>Supporting infrastructure improvements to better track or diagnose gameplay related issues.</li></ul></div><div>Senior Machine Learning Engineer</div><div>January, 2021 – 2024</div><div><ul style="list-style-type: none"><li>Designed, built, maintained, and improved ML infrastructure. This includes tool developement like autoML, CI/CD (Jenkins, github actions), model tracking and management (MLflow) and orchestration (Airflow).</li><li>Crafted and implimented policy around external data ingestion and data product retention for both my internal team and for sharing across the organization.</li><li>Designed, built, and maintained near-realtime applications (via Spark streaming) to combat in-game cheating and other malicious behavior. Typical time-to-action in the low 10s of seconds.</li><li>Lead the transition of ML infrastructure from AWS to GCP/GCS.</li><li>Created ML models to provide insights into customer conversion, churn, and behavioral segmentation. This leveraged survival analysis, clustering, as well as tree-based and linear methods.</li><li>Supervised junior team members to design and developed recommendation systems productionalized in both Modern Warfare 2 (2022) and Modern Warfare 3 (2023).</li></ul></div><div>Insight Data Science</div><div>New York, New York</div><div>Fellow</div><div>January, 2020 – 2021</div><div><ul style="list-style-type: none"><li>Helped optimize the way NYC health inspectors perform restaurant inspections in order to reduce the time critical health violations remain unaddressed.</li><li>Trained a random forest in Python to prioritize NYC restaurant inspections based on environmental variables and their past inspection histories and provided the results to NYC through an API deployed on AWS.</li><li>Resulted in NYC inspectors identifying ~2.5% more violations in the first half of an inspection window, leading to critical violations being discovered up to 7 days earlier than by the current approach implemented by NYC.</li></ul></div><div>Dept. of Physics and Astronomy, Rutgers University</div><div>New Brunswick, New Jersey</div><div>Postdoctoral Research Associate</div><div>September, 2016 – 2021</div><div><ul style="list-style-type: none"><li>Designed and built parallelized pipelines to process and analyze TBs of astronomical imaging; producing calibrated, standardized data catalogs and rigorous results leading to 2 peer reviewed publications and several hundred hours of telescope time.</li><li>Contributed to open source, astronomy-focused, Python projects through bug fixes and feature additions: see photometrypipeline, astLib, and easyGalaxy on GitHub as examples.</li></ul></div></div> |  |
| Skills                  | <div>Machine Learning: Linear Models, Decision Trees, SVM, Clustering, Deep Learning, Survival Analysis</div> <div>CI/CD: Jenkins, Airflow, Docker</div> <div>Software and Computing: Open Source Contributor, Python, DataBricks, MLFlow, SQL, AWS/GCP, Spark, and other cloud computing applications</div> <div>Leadership: Experience organizing and leading workshops and collaboration meetings, Teaching and mentoring junior team members, Eagle Scout.</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
| Education               | <div>Texas A&amp;M University, College Station, Texas</div> <div>The University of Tennessee, Knoxville, Tennessee</div> <div><ul style="list-style-type: none"><li>Ph.D, Physics (astronomy focus), 2016</li><li>M.S., Physics (astronomy focus), 2009</li><li>B.S., Physics, 2007</li></ul></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |